Vansh Gupta

+91 9289011325 | vanshgupta7266@gmail.com | LinkedIn | GitHub
CodeChef **4-Star** (max 1856) | Codeforces **Pupil** (max 1340) | Bhopal, India

Professional Summary

Third-year B.Tech student at IIIT Bhopal (CGPA: **8.49/10**) with 2+ years of programming experience in Python and C++, applying engineering principles to design, develop, and test production-grade software systems — from agentic AI pipelines to reinforcement learning environments and ML forecasting platforms. Demonstrates strong computer science fundamentals — data structures, algorithms, and object-oriented design — by solving complex problems through sound and creative engineering, and through active competitive programming (CodeChef **4-Star** | Codeforces **Pupil**). Currently an ML Intern at Moleculyst, quickly adopting new engineering methods across multimodal AI and applied quantitative research, reviewing current developments and seeking new knowledge to improve the reliability, efficiency, and performance of production systems. A collaborative, feedback-driven engineer who applies best practices to improve technical solutions, delivers software projects on schedule, and thrives in cooperative team environments — committed to high availability, observability, and consistent operations at scale.

Education

IIIT Bhopal **Bhopal, Madhya Pradesh, India**
B.Tech in Information Technology (CS & Engineering Curriculum) | *CGPA: **8.49 / 10*** *Aug 2024 – May 2028*

- **Coursework:** Computer Science Fundamentals, Data Structures & Algorithms, Object-Oriented Programming (OOP), Operating Systems, Database Management Systems, Computer Networks, Discrete Mathematics

Technical Skills

OOP Languages: Python, C++, C, SQL

CS Fundamentals: Data Structures, Algorithms, Object-Oriented Design, Time/Space Complexity Analysis, Algorithmic Reasoning

Software Engineering: End-to-End Software Design & Development, Software Testing & Debugging, REST API Development (FastAPI), Version Control (Git/GitHub), Improving Technical Solutions, On-Schedule Project Delivery

System Reliability & Performance: Availability, Reliability, Efficiency, Observability, Performance, Monitoring, Operations at Scale, Scalable Pipeline Architecture, Cloud Deployment (AWS, GCP / Cloud Run, Vertex AI)

Competitive Programming (C++): CodeChef 4-Star (max **1856**) | **Codeforces Pupil** (max **1340**) — DP, graph algorithms, segment trees, combinatorics

AI / ML Stack: PyTorch, TensorFlow, Scikit-learn, LangChain, LangGraph, Docker, LLMs / VLMs, RAG, Reinforcement Learning, LSTM, XGBoost

Experience

ML Intern **Feb 2026 – Present**
Moleculyst *Remote*

- Applied engineering principles to build multimodal Video LLM inference pipelines in Python — integrating vision-language architectures, preprocessing stages, and structured evaluation loops — delivering Moleculyst's core inference system for structured understanding across complex operational video datasets.
- Improved pipeline reliability and efficiency by designing structured evaluation frameworks that standardized quality assessment across model versions, enabling reproducible, consistent benchmarking without manual intervention.
- Reviewed current engineering developments and adopted new engineering methods by expanding scope into applied quantitative research — incorporating probability theory, mathematical optimization, and statistical modeling — seeking new technical knowledge to meet Moleculyst's evolving engineering requirements.
- Collaborated with appropriate stakeholders to determine user requirements for features, iterate on technical designs, and apply best practices to improve technical solutions — delivering software projects on schedule in a cooperative team environment.

Projects

Equitas AI | *Python, LangGraph, FastAPI, Gemini API, Vertex AI, GCP* **2025**

- Designed a **5-agent** autonomous LangGraph pipeline for dataset bias auditing, orchestrating iterative mitigation loops that converged at Disparate Impact Ratio (DIR) $\geq$ **0.80** — the U.S. legal fairness threshold — across all evaluated datasets.
- Improved interpretability for non-technical stakeholders by integrating SHAP explainability matrices with Gemini-driven counterfactual narration, surfacing row-level bias reasoning through an interactive review interface.
- Eliminated credential management overhead by deploying a keyless FastAPI backend on Google Cloud Run via Vertex AI — delivering a fully serverless production system with autoscaling availability and no stored secrets.

Triage AI | *Python, FastAPI, OpenEnv, Pydantic, Docker, LLaMA 3.1* **2025**

- Architected a reinforcement learning environment for automated email classification by implementing structured OpenEnv state-machine cycles (reset / step / observe) with multi-level prompt engineering chains, achieving **0.71 / 1.0** reward accuracy on LLaMA 3.1 (8B).
- Containerized the full system with Docker and enforced API contracts with Pydantic — producing a reproducible, environment-isolated deployment that can be reliably reset and re-evaluated across test cycles.

PharmAI | *Python, LSTM, XGBoost, SHAP, Scikit-learn* **2025**

- Engineered a pharmaceutical batch quality prediction system using LSTM and XGBoost models trained on historical process data, achieving **>90% accuracy** on held-out validation batches.
- Reduced defect risk by implementing SHAP-driven anomaly attribution for early failure detection, surfacing the top contributing process variables per flagged batch for corrective action.

Moleculyst Studio | *Python, Gemini API, VLMs, OpenCV, Docker* **2025**

- Built a multi-step agentic AI editing workspace composing VLM-based grounding, targeted edit operators, and an automated critique loop — replacing manual step-by-step review with a single-command, end-to-end pipeline.
- Improved operational efficiency and debuggability by streaming structured reasoning traces, tool calls, and per-step quality scores into a unified monitoring view — enabling engineers to inspect and intervene at any stage of the pipeline.

Achievements

- **Meta × PyTorch OpenEnv Hackathon — Grand Finalist** (Bangalore, 2025): Top-tier ranking out of **52,000+** global participants for shipping a production-ready reinforcement learning environment.
- **CodeBidz Hackathon — 1st Place:** Won as a solo developer, independently executing the full product lifecycle — architecture, engineering, and delivery — under competition time constraints.
- **OpenAI × Outskill AI Builders Hackathon:** Shortlisted in **Top 1,000** developers nationwide to design, optimize, and ship a production-grade live AI product.
- **JEE Main 2024 — 98.16 percentile:** Top 1% nationally out of 1.2M+ candidates; qualified for JEE Advanced.
- **SRMC Mathematics Competition — AIR 27 (Finals):** National finalist in the tournament jointly organized by IIT Bombay & IIT Madras.